

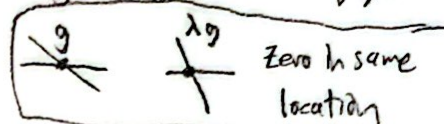
Comments on Lagrange multipliers

Cost: $f(x)$ Constraints: $g_i(x)=0$ x is a vector.

We want a function with a gradient that vanishes on $g(x)=0$ and sits on f .

$\mathcal{L} = f + g$ fits this condition, since $f + g = f$ when $g = 0$. To make the gradient vanish, take $\nabla \mathcal{L} = \nabla f + \nabla g = 0$, $\nabla f = -\nabla g$, $\nabla f = -\lambda \nabla g$. Add a λ so everything cancels accordingly.

Note that $g=0$ and $\lambda g=0$ still refer to the same constraint.



In order for $\nabla f = -\lambda \nabla g$ to be true, the extreme points of f and $-\nabla g$ should be on opposite sides of the constraint.



The gradients that cancel form a straight line between the two extreme points. Therefore, the closest possible point to f^* .

* gradients are always perpendicular to contours, b/c gradients are 0 along contours.

* the gradients of f and g line up when a contour of f is tangent to the constraint g .

This is because at an extreme point there is only one spot it appears. Otherwise it's not an extreme point.



One value of f must be a contour, and moving to either side of that contour only moves you up or down.

* Not much stopping you from flipping the roles of f and g .

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